Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In Young's double slit experiment, the fringe width is $2.0\ \text{mm}$, The separation between the 9 th bright fringe and the second dark fringe from the centre of the fringe system will be

(A) $5.0\ \text{mm}$

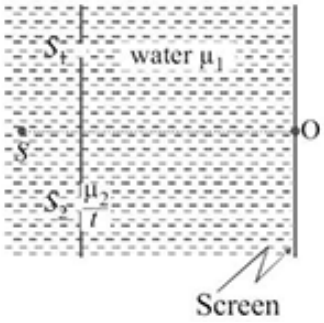
(B) $10\ \text{mm}$

(C) $15\ \text{mm}$

(D) $20\ \text{mm}$

Q.2

A YDSE is conducted in water (μ_1) as shown in figure. A glass plate of thickness t and refractive index μ_2 is placed in the path of S_2. The optical path difference at O is



(A) $(\mu_2 - 1)\, t$

(B) $(\mu_1 - 1)\, t$

(C) $\left(\frac{\mu_2}{\mu_1} - 1\right) t$

(D) $(\mu_2 - \mu_1)\, t$

Q.3

Two sources of light of wavelengths $2500\mathring{A}$ and $3500\mathring{A}$ are used in Young's double slit expt. simultaneously. Which orders of fringes of two wavelength patterns coincide?

(A) 3rd order of 1st source and 5 th of the 2 nd

(B) 7 th order of 1 st and 5 th order of 2 nd

(C) 5 th order of 1 st and 3rd order of 2 nd

(D) 5 th order of 1 st and 7 th order of 2 nd

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Q.4

The intensity at the maximum in a Young's double slit experiment is I_0. Distance between two slits is $d = 5l$, where 1 is the wavelength of light used in the experiment. What will be the intensity in front of one of the slits on the screen placed at a distance $D = 10\ d$?

(A) I_0

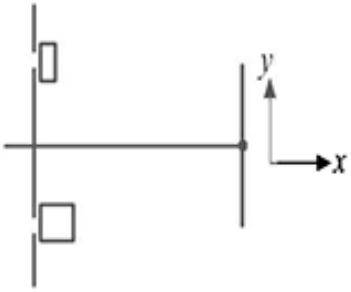
(B) $\frac{I_0}{4}$

(C) $\frac{3}{4}I_0$

(D) $\frac{I_0}{2}$

Q.5

In YDSE, both slits are covered by transparent slab. Upper slit is covered by slab of R.I. 1.5 and thickness t and lower is covered by R.I. $\frac{4}{3}$ and thickness $2t$, then central maxima



(A) shifts in +ve

(B) shifts in -ve y

(C) remains at

(D) may shift in upward or downward depending upon wavelength of light

Q.6

In the Young's double-slit experiment, the intensity of light at a point on the screen where the path difference is λ is K, (λ being the wave length of light used). The intensity at a point where the path difference is $\lambda/4$, will be:

(A) K

(B) $K/4$

(C) $K/2$

(D) Zero

Q.7

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If yellow light emitted by sodium lamp in Young's double slit experiment is replaced by a monochromatic blue light of the same intensity

(A) fringe width will decrease

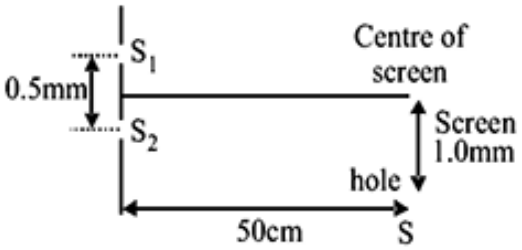
(B) finge width will increase

(C) fringe width will remain unchanged

(D) fringes will become less intense

Q.8

In Young's double slit experiment shown in figure S_1 and S_2 are coherent sources and S is the screen having a hole at a point $1.0\ mm$ away from the central line. White light $(400$ to $700\ nm)$ is sent through the slits. Which wavelength (in nm) passing through the hole has strong intensity?



(A) 400nm

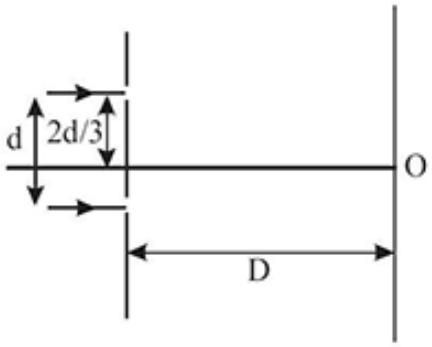
(B) 700nm

(C) 500nm

(D) 667nm

Q.9

In the figure shown if a parallel beam of white light is incident on the plane of the slits then the distance of the nearest white spot on the screen from
O is d/A. Find the value of A. (assume $d << D, \lambda \ll d]$



(A) 6

(B) 3

(C) 4

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(D) 1

Q.10

A double slit arrangement produces fringes for $\lambda = 5890\mathring{A}$ that are 0.4° apart. What is the angular width if the entire arrangement is immersed in water ? $(\mu_w = 4/3)$

(A) 0.3°

(B) 2.3°

(C) 0.8°

(D) 1.3°

Q.11

Young's double slit experiment is made in a liquid. The 10th bright fringe lies in liquid where 6th dark fringe lies in vacuum. The refractive index of the liquid is approximately

(A) 1.8

(B) 1.5

(C) 1.3

(D) 1.6

Q.12

Interference fringes were produced in Young's double slit experiment using light of wavelength $5000\mathring{A}$. When a film of material 2.5×10^{-3} cm thick was placed over one of the slits, the fringe pattern shifted by a distance equal to 20 fringe width. The refractive index of the material of the film is

(A) 1.25

(B) 1.33

(C) 1.4

(D) 1.5

Q.13

Two beams of light having intensities I and $4I$ interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\pi/2$ at point A and π at point B. Then the difference between the resultant intensities at A and B is

(A) $2I$

(B) $4I$

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(C) $5I$

(D) $7I$

Q.14

In the ideal double-slit experiment, when a glassplate (refractive index 1.5) of thickness t is introduced in the path of one of the interfering beams (wave-lenght λ), the intensity at the position where the central maximum occurred previously remains unchanged. The minimum thickness of the glass-plate is

(A) 2λ

(B) $2\lambda/3$

(C) $\lambda/3$

(D) λ

Q.15

Monochromatic light of wavelength $400\ \mathrm{nm}$ and $560\ \mathrm{nm}$ are incident simultaneously and normally on double slits apparatus whose slits separation is $0.1\ \mathrm{mm}$ and screen distance is $1\ \mathrm{m}$. Distance between areas of total darkness will be

(A) $5.6\ \mathrm{mm}$

(B) $4\ \mathrm{mm}$

(C) $14\ \mathrm{mm}$

(D) $28\ \mathrm{mm}$

Q.16

White light used to illuminate the two slits in Young's double slit experiment. The separation between the slits is d and the distance between the screen and the slit is $\mathrm{D}(>> \mathrm{d})$. At a point on the screen in front of one of the slits, certain wavelengths are missing. The missing wavelengths are

(A) $\lambda = \frac{\mathrm{d}^2}{(2\mathrm{n}+1)\mathrm{D}}$

(B) $\lambda = \frac{(2\mathrm{n}+1)\mathrm{d}^2}{\mathrm{D}}$

(C) $\lambda = \frac{\mathrm{d}^2}{(\mathrm{n}+1)\mathrm{D}}$

(D) $\lambda = \frac{(\mathrm{n}+1)\mathrm{D}}{\mathrm{d}^2}$

Q.17

Interference fringes were produced using white light in a double slit arrangement. When a mica sheet of

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uniform thickness of refractive index 1.6 (relative to air) is placed in the path of light from one of the slits, the central fringe moves through some a distance. This distance is equal to the width of 30 interference bands if light of wavelength 4800 $\mathring{A}$ is used. The thickness (in $\mu \mathrm{m}$) of mica is

(A) 90

(B) 12

(C) 14

(D) 24

Q.18

In a Young's double slit experiment, 12 fringes are observed to be formed in a certain segment of the screen when light of wavelength $600 \mathrm{~nm}$ is used. If the wavelength of light is changed to $400 \mathrm{~nm}$, number of fringes observed in the same segment of the screen is given by

(A) 12

(B) 18

(C) 24

(D) 30

Q.19

In Young's double slit experiment intensity at a point is $(1/4)$ of the maximum intensity. Angular position of this point is

(A) $\sin^{-1}(\lambda/d)$

(B) $\sin^{-1}(\lambda/2 \mathrm{~d})$

(C) $\sin^{-1}(\lambda/3 \mathrm{~d})$

(D) $\sin^{-1}(\lambda/4 \mathrm{~d})$

Q.20

When a plastic thin film of refractive index 1.45 is placed in the path of one of the interfering waves then the central fringe is displaced through width of five fringes. The thickness of the film, if the wavelength of light is $5890 \mathring{A}$, will be

(A) $6.544 \times 10^{-4} \mathrm{~cm}$

(B) $6.544 \times 10^{-4} \mathrm{~m}$

(C) $6.54 \times 10^{-4} \mathrm{~cm}$

(D) $6.5 \times 10^{-4} \mathrm{~cm}$

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Q.1 Answer:

15 mm

Solution:

The distance of the nth bright fringe from the central fringe is

$y_n = n\lambda\frac{D}{d} = n\beta$

where $\beta = \frac{\lambda D}{d}$ is the fringe width.

$\therefore y_n = 9\beta$ **(1)**

The distance of the m th dark fringe from the central fringe is

$y_n' = \left(m - \frac{1}{2}\right)\frac{\lambda D}{d} = \left(m - \frac{1}{2}\right)\beta$

$y_2' = \frac{3}{2}\beta$

From Eqs. (1) and (2) we get

$$y_9 - y_2' = 9\beta - \frac{3}{2}\beta = \frac{15}{2}\beta$$

$$= \frac{15}{2} \times 2.0 \text{ mm} = 15 \text{ mm}$$

Hence the correct choice is (c).

Q.2 Answer:

$\left(\frac{\mu_2}{\mu_1} - 1\right)t$

Solution:

Correct option is (a) $\left|\left(\frac{\mu_2}{\mu_1} - 1\right)\right|$

When no slab is present in front of slit 2 $\Rightarrow S_1 = S_2$

When a slab is placed in front of slit, some deflection occurs which is equals to the optical path difference.

$$\therefore \Delta x = \left(\frac{\mu_2}{\mu_1} - 1\right)$$

Optical path difference at $O = \left(\frac{\mu_2}{\mu_1} - 1\right)$

Q.3 Answer:

7 th order of 1 st and 5 th order of 2 nd

Solution:

Let nth fringe of $2500\mathring{A}$ **coincide with** $(n-2)$ **th fringe of** $3500\mathring{A}$

$\therefore 3500(n-2) = 2500 \times n$

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$1000n = 7000, n = 7$

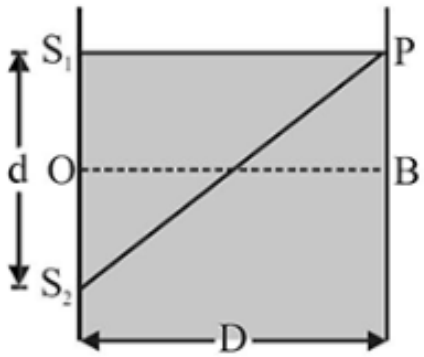
$\therefore$ 7 th order fringe of 1 st source will coincide with 5 th order fringe of 2 nd source.

Q.4 Answer:

$\frac{I_0}{2}$

Solution:

Let P is a point infront of one slit at which intensity is to be calculated. From figure,



Path difference $= S_2P - S_1P$

$= \sqrt{D^2 + d^2} - D = D\left(1 + \frac{1}{2}\frac{d^2}{D^2}\right) - D$

$= D\left[1 + \frac{d^2}{2D^2} - 1\right] = \frac{d^2}{2D}$

$\Delta x = \frac{d^2}{2 \times 10d} = \frac{d}{20} = \frac{5\lambda}{20} = \frac{\lambda}{4}$

Place difference,

$\Delta\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$

So, resultant intensity at the desired point ' p ' is $I = I_0 \cos^2 \frac{\phi}{2} = I_0 \cos^2 \frac{\pi}{4} = \frac{I_0}{2}$

Q.5 Answer:

shifts in -ve y

Solution:

$\Delta x_1 = (\mu_1 - 1)\, t = (1.5 - 1)t = 0.5t$

and $\Delta x_2 = (\mu_2 - 1) \times 2t = \left(\frac{4}{3} - 1\right) \times 2t = \frac{2}{3}t.$

As $\Delta x_2 > \Delta x_1$**, so shift will be along** $-$ **ve** y **-axis.**

Q.6 Answer:

$K/2$

Solution:

For path difference λ**, phase difference** $= 2\pi$rad.

For path difference $\frac{\lambda}{4}$**, phase difference** $= \frac{\pi}{2}$ **rad. As** $K = 4I_0$ **so intensity at given point where path**

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difference is $\frac{\lambda}{4}$

$K' = 4I_0 \cos^2\left(\frac{\pi}{4}\right)\left(\cos\frac{\pi}{4} = \cos 45^\circ\right)$

$= 2I_0 = \frac{K}{2}$

Q.7 Answer:

fringe width will decrease

Solution:

As $\beta = \frac{\lambda D}{d}$ and $\lambda_b < \lambda_y$,

$\therefore$ fringe width β will decrease

Q.8 Answer:

500nm

Solution:

Wavelength for which maximum obtained at the hole has the maximum intensity on passing. So,

$x = \frac{n\lambda D}{d}$

$\lambda = \frac{xd}{nD} = \frac{1\times10^{-3}\times0.5\times10^{-3}}{n\times50\times10^{-2}}$

$= \frac{1\times10^{-6}}{n} = \frac{1000\text{ nm}}{n}$

$n = 1, \lambda = 1000\text{ nm} \rightarrow$ **Not in the given range**

$n = 2, \lambda = 500\text{ nm}$

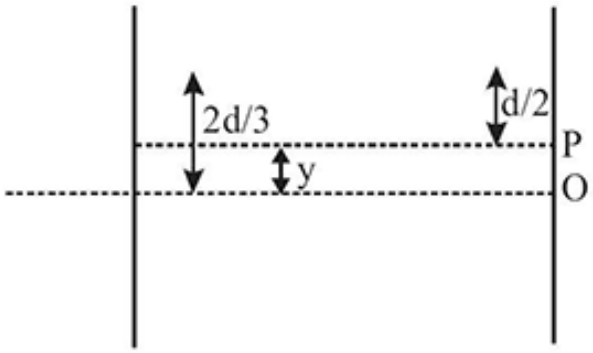
Q.9 Answer:

6

Solution:

The nearest white spot will be at P, the central maxima.

$\therefore y = \frac{2d}{3} - \frac{d}{2} = \frac{d}{6}$



Q.10 Answer:

0.3°

Solution:

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We know the distance between two slits $\mathrm{d} = \frac{\lambda}{\sin\theta}$, **where** $\lambda =$ **wavelength and** θ **= angle from the original beam.**

Given wavelength of sodium light $\lambda_1 = 5890\mathring{A} = 5890 \times 10^{-10}$ m **and** $\theta_1 = 0.40^\circ$
Putting the above values in the given equation we get,

$$\mathrm{d} = \frac{5890 \times 10^{-10}}{\sin 0.40} = 8.43 \times 10^{-5} \text{ m}$$

Also $\lambda_2 = \frac{\lambda_1}{\mathrm{n}} \ldots - (\mathrm{A})$
where $\mathrm{n} =$ **refractive index of water** $= 1.33$
Thus A becomes,

$$\lambda_2 = \frac{5890 \times 10^{-10}}{1.33} = 4.42 \times 10^{-7} \text{ m}$$

Again,

$$\mathrm{d} = \frac{\lambda_2}{\sin\theta_2}$$
$$\theta_2 = \sin^{-1}\left(\frac{4.42\times10^{-7}}{1.33}\right) = 0.30^\circ$$

Q.11 Answer:

1.8

Solution:

In liquid, wavelenth reduces by a factor equal to refractive index of liquid μ**. Path difference is equal for bright and dark condition**

$$\therefore 10\frac{\lambda}{\mu} = (2 \times 6 - 1)\frac{\lambda}{2}$$
$$\mu = \frac{20}{11} = 1.81$$

Q.12 Answer:

1.4

Solution:

$n = \frac{(\mu-1)tD}{d}$ **but** $\beta = \frac{\lambda D}{d} \Rightarrow \frac{D}{d} = \frac{\beta}{\lambda}$
$n = (\mu - 1)t\beta/\lambda$

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$20\beta = (\mu - 1)2.5 \times 10^{-3}\left(\beta/5000 \times 10^{-8}\right)$

$\mu - 1 = \frac{20 \times 5000 \times 10^{-8}}{2.5 \times 10^{-3}} \Rightarrow \mu = 1.4$

Q.13 Answer:

$4I$

Solution:

When Two beams of light having intensities I_1 and I_2 interfere and produce a fringe pattern on a screen, the resultant intensity, $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\varphi$, φ is the phase difference.

For point A, two beams of light having intensities I and $4I$ interfere,

Hence, the resultant intensity will be $I_A = I + 4I + 2\sqrt{I.4I}\cos\varphi$

The phase difference at point A, $\varphi = \frac{\pi}{2}$

$$\therefore I_A = 5I + 2\sqrt{4I^2}\cos\tfrac{\pi}{2}$$
$$\Rightarrow I_A = 5I + 4I \times 0 \text{ since }, \$\cos\tfrac{\pi}{2} = 0\$$$
$$\Rightarrow I_A = 5I$$

For point B, two beams of light having intensities I and $4I$ interfere

Hence, the resultant intensity will be $I_B = I + 4I + 2\sqrt{I.4I}\cos\varphi$

The phase difference at point B, $\varphi = \pi$

$$\therefore I_B = 5I + 2\sqrt{4I^2}\cos\pi$$
$$\Rightarrow I_B = 5I + 4I(-1) \text{ since }, \$\cos\pi = -1\$$$
$$\Rightarrow I_B = 5I - 4I$$
$$\Rightarrow I_B = I$$

Therefore, the difference between the two resultant intensities is

$$I_A - I_B = 5I - I = 4I$$

Q.14 Answer:

2λ

Solution:

Path difference $= (\mu - 1)\mathrm{t} = \mathrm{n}\lambda$

For minimum $t, n = 1; \therefore t = 2\lambda$

Q.15 Answer:

$28\ \mathrm{mm}$

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Solution:

At the area of total darkness minima will occur for both the wavelengths.

$\therefore \quad \frac{(2n+1)}{2}\lambda_1 = \frac{(2m+1)}{2}\lambda_2$

$\Rightarrow \quad (2n+1)\lambda_1 = (2m+1)\lambda_2$

or $\frac{(2n+1)}{(2m+1)} = \frac{560}{400} = \frac{7}{5}$ **or** $10n = 14m + 2$

by inspection for $m = 2, n = 3$ **and for** $m = 7, n = 10$**, the distance between them will be the distance between such points.**

i.e., $\Delta s = \frac{D\lambda_1}{d}\left\{\frac{(2n_2+1)-(2n_1+1)}{2}\right\}$

put $n_2 = 10, n_1 = 3$

On solving we get, $\Delta s = 28$ mm.

Q.16 Answer:

$\lambda = \frac{\mathrm{d}^2}{(2\mathrm{n}+1)\mathrm{D}}$

Solution:

n^{th} **minimum has a distance from the centre** $= x = (2n-1)\frac{1}{2}\frac{\lambda D}{d}$

For a point on the screen directly in front of one of the slits, $\mathrm{x} = \mathrm{d}/2 \therefore$ **for minimum intensity in front of of the slits**

$= \frac{\mathrm{d}}{2} = (2\mathrm{n}-1)\frac{\lambda}{2}\frac{\mathrm{D}}{\mathrm{d}} \therefore \lambda = \frac{\mathrm{d}^2}{(2\mathrm{n}-1)\mathrm{D}}$

Q.17 Answer:

24

Solution:

Shift of fringe pattern $= (\mu - 1)\frac{tD}{d} \therefore \frac{30D\left(4800\times10^{-10}\right)}{d} = (0.6)t\frac{D}{d}$

$30 \times 4800 \times 10^{-10} = 0.6t$

$t = \frac{30\times4800\times10^{-10}}{0.6} = \frac{1.44\times10^{-5}}{0.6} = 24 \times 10^{-6}$

Q.18 Answer:

18

Solution:

$\frac{12\lambda_1 D}{d} = \frac{k\lambda_2 D}{d}, k = \frac{12\times600}{400} = 18$

Q.19 Answer:

$\sin^{-1}(\lambda/3\ \mathrm{d})$

Solution:

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Correct option is (c) $\sin^{-1}\left(\frac{\lambda}{3\,\mathrm{d}}\right)$

Interesting at a point (I) where phase difference is δ **bx Y DSE in given by :** $\mathrm{I} = \mathrm{Imax}\cos^2\delta/2$ **[Imax: maximum intensity]**

at the point where $\mathrm{I} = \frac{\mathrm{Imax}}{4}$**, we get :**

$$\frac{\mathrm{Imax}}{4} = \mathrm{Imax}\cos^2\delta/2$$
$$\Rightarrow \cos^2\delta/2 = \frac{1}{4} \Rightarrow \cos\delta/2 = \pm\frac{\delta}{2} = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}$$

$$\Rightarrow \delta = 2\frac{\pi}{3} \text{ or } \frac{2\pi}{3}, \text{ well take } \frac{2\pi}{3} \text{ only.}$$

Thus path difference $(\triangle x)$ **at that point in,**

$$\Delta\mathrm{x} = \frac{\lambda}{2\pi}\delta = \frac{2\lambda}{2\pi}\cdot\frac{\pi}{3}. \Rightarrow \mathrm{x} = \frac{\lambda}{\mathrm{B}}$$

Also, $\triangle\mathrm{x} = \mathrm{d}\sin\theta(\theta$ **: angular position)**

$$\Rightarrow \sin\theta = \frac{\lambda}{3\,\mathrm{d}} \Rightarrow \theta = \sin^{-1}\left(\frac{\lambda}{3\,\mathrm{d}}\right)$$

Q.20 Answer:

6.544×10^{-4} cm

Solution:

$\therefore X_0 = \frac{\beta}{\lambda}(\mu - 1)t \Rightarrow 5\beta = \frac{\beta(0.45)t}{5890\times10^{-10}}$
$\therefore t = \frac{5\times5890\times10^{-10}}{0.45} = 6.544 \times 10^{-4}$ cm

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